

Long-Context Memory

All tokens stay live in the context window.

When Transient

Where Internal

What Raw observations

How Write; Consolidate

Representative papers

VideoBERT · HERO · LongVILA · Video-XL

Compressed Visual Memory

Squeezes frames into a few latent tokens.

When Transient

Where Internal

What Latent Representations

How Write; Consolidate

Representative papers

Flamingo · BLIP-2 · TokenLearner · MovieChat

KV-Cache Memory

Reuses attention key-values on a budget.

When Transient

Where Internal

What Latent representations

How Write; Consolidate; Update & Forget

Representative papers

LOOK-M · MadaKV · AKVQ-VL

Session History Memory

Logs the session as turns, slots, or steps.

When Episodic

Where Internal

What Structured Episodic Records

How Write; Consolidate

Representative papers

VD-BERT · Multimodal DST · RecFormer · ETPNav

Recurrent Memory

Carries a rolling latent state forward.

When Episodic

Where Internal

What Latent representations

How Write; Consolidate

Representative papers

MART · MA-LMM · VideoLLaMB

External Retrieval-Augmented Memory

Looks up an external knowledge store.

When Persistent

Where External

What Abstract semantic knowledge

How Write; Consolidate; Retrieve

Representative papers

REALM · RAG · MuRAG · Reveal · VisRAG

In-Weights Parametric Memory

Bakes knowledge into the model parameters.

When Persistent

Where External

What Abstract semantic knowledge

How Update & Forget

Representative papers

LwF-VLM · LLaMA-Adapter · Can-We-Edit

Lifelong Agent Memory

Builds up experience across many sessions.

When Persistent

Where External

What Structured episodic records;
Abstract semantic knowledge

How Write; Consolidate; Retrieval;
Update & Forget

Representative papers

MemoryBank · KARMA · Voyager · AtlasVA